Employee Salary & Compensation Estimator

Machine Learning + Streamlit Project Report

Objective: Predict annual employee compensation using employee experience, education, role, geography/city tier and related attributes, then provide an interactive salary benchmark.

Dataset: employee_salary_regression.csv — 1,000 records, 10 columns, no missing values and no duplicate rows.

Target: annual_salary_usd

Preprocessing: employee_id is excluded; categorical variables use One-Hot Encoding; numeric variables use StandardScaler; all preprocessing is contained inside the model Pipeline.

Models compared: Linear Regression, Random Forest Regression and Gradient Boosting Regression.

Model Evaluation

Model	MAE	RMSE	R^2
Linear Regression	$3,524.83	$4,381.87	0.9854
Random Forest	$4,843.44	$5,972.64	0.9729
Gradient Boosting	$4,157.91	$5,068.40	0.9805

Best model: Linear Regression

Test R^2: 0.9854

Test MAE: $3,524.83

Streamlit application features include employee input sliders/selectors, salary prediction, role-level market average, observed salary range, salary position, employee profile summary, model comparison and dataset overview.

Important dataset provenance note: The supplied project ZIP did not contain the original Kaggle URL or download metadata. If the instructor requires proof of Kaggle provenance, the exact Kaggle dataset URL must be added before submission.

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Run command: python -m streamlit run app.py
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